



Phase 0A: Compliance Audit & Product Design

Cross-Regional Regulatory Analysis — AU · US · NZ · UK

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Classification: Commercial in Confidence

15 Work Items · 4 Jurisdictions · 8 Documents

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1. Executive Summary

FutureProof's Equity Participation Mortgage (EPM) has been assessed across four target jurisdictions: Australia, the United States, New Zealand, and the United Kingdom. This audit covers regulatory classification, licensing requirements, tax treatment, estate planning, insurance obligations, security posture, and incident response readiness.

15

Work Items Completed

4

Jurisdictions Analysed

~\$400K

One-Time Licensing Cost

12 mo

Critical Path to Launch

Key Conclusion

EPM is viable in all four regions, but product structure is paramount.

Model B (lender owns portfolio, consumer receives loan advances)

provides the optimal tax, regulatory, and consumer outcome across every jurisdiction. This single decision eliminates FIF tax (NZ), Centrelink deeming (AU), dividend tax (UK), and 1099 reporting (US).

Top 5 Findings

#	Finding	Impact	Priority
1	Model B (lender owns portfolio) recommended for all regions	Tax-free income everywhere; simplifies consumer experience	Decision
2	US: SEC risk — Howey test arguably met	Must structure as loan advances, not investment returns. Engage securities counsel.	Critical
3	UK: Lifetime mortgage classification — advised sales only	No self-serve. ERC membership essential. NNEG mandatory.	Critical
4	AU: Centrelink impact — Age Pension likely reduced	\$640K portfolio exceeds assets test. Must disclose prominently.	High
5	Security gaps — No MFA, no field encryption, InputSanitization disabled	Must be remediated before launch.	Critical

2. Critical Design Decisions

These six decisions were identified during the audit as having the highest impact on product viability. They affect every aspect of the platform — from financial modelling to UX to legal documentation.

Decision 1: Portfolio Ownership — Model B

Recommendation: Lender/SPV owns the investment portfolio. Consumer receives loan advances only.

This is the single most consequential product design decision. It determines tax treatment, regulatory classification, consumer experience, and estate outcomes in all four regions.

Factor	Model A (Consumer Owns)	Model B (Lender Owns)
Income tax on EPM payments	Potentially taxable as investment income	Not taxable (loan proceeds)
NZ FIF tax	~\$10,560/year on \$640K portfolio	\$0 — lender entity bears FIF
AU Centrelink deeming	Portfolio deemed to earn income	Reduced exposure (loan, not investment)
US 1099 reporting	Annual 1099-DIV, 1099-B	None — no taxable distributions
UK dividend tax	Dividend allowance then 8.75-39.35%	\$0 — no dividends to consumer
Consumer tax compliance	Annual tax return complexity	None
Net income to consumer	Higher gross, lower after tax	Lower gross, higher after tax

Decision 2: Advised Sales Only

All regions require or strongly favour advised sales. No self-serve EPM purchase flow should be built.

- **UK:** FCA requires advice for lifetime mortgage products
- **AU:** AFSL likely required — personal advice obligations apply
- **NZ:** FAP licence conditions require competent advice
- **US:** HUD counseling required for reverse-mortgage-adjacent products

Decision 3: NNEG is Non-Negotiable

Every EPM contract must include a No Negative Equity Guarantee. The consumer (and estate) will never owe more than the property is worth at settlement.

- UK: Mandatory (ERC membership requirement)
- US: Standard for HECM (HUD requirement)
- AU/NZ: Contractual commitment (not yet regulated for EPM, but commercially essential)

Decision 4: CPI-Linked Income

Annual income adjustment by local CPI (capped at 4% p.a.). Sustainable for 25+ years based on modelling. Without inflation adjustment, income loses ~45% of purchasing power over 20 years.

Decision 5: Security Hardening (Before Launch)

6 critical security gaps must be closed before production launch:

Re-enable InputSanitization on User model · Enable Devise :lockable · Enable Devise :confirmable · Implement MFA for admin · Field-level encryption for L4 data · Strengthen passwords (12+ admin, 10+ user)

Decision 6: Regional Disclosures

Region	Mandatory Disclosure
AU	Centrelink impact estimate in every quote
US	"Loan proceeds are not taxable income"
NZ	"Does not affect NZ Superannuation"
UK	IHT impact calculator in every quote
All	"Seek independent tax/financial advice" at quote, application, and contract

3. Regional Compliance Overview

Regulatory Classification

Region	Likely Classification	Primary Licence	Key Risk
Australia	Credit product (NCC) + possibly financial product	ACL + potentially AFSL	ASIC dual classification
United States	Mortgage loan (non-HECM)	NMLS + state licences	SEC securities classification (Howey test)
New Zealand	Credit contract (CCCFA)	FSP + FAP	FIF tax on offshore ETFs
United Kingdom	Lifetime mortgage (MCOB 3A/9A)	FCA authorisation	Advised sales only; ERC membership

Consumer Protection Comparison

Protection	AU	US	NZ	UK
Cooling-off period	✔ 5 days	✔ 3 days (TILA)	✔ 5 days	✔ 14 days
Right to independent advice	✔ Recommended	✔ HUD counseling	✔ Required	✔ Required (FCA)
NNEG / Non-recourse	⚠ Contractual	✔ FHA standard	⚠ Contractual	✔ ERC mandatory
Dispute resolution	AFCA	CFPB + state AG	FDRS/FSCL	FOS
Breach notification	30 days	30-90 days (state)	~72 hours	72 hours (GDPR)

EPM's Unique Regulatory Position

EPM is a consumer zero-risk product.

The lender bears ALL investment risk, market risk, and longevity risk. The consumer receives guaranteed income, retains their home, and is protected by NNEG. This is fundamentally different from traditional reverse mortgages where compound interest erodes the estate.

4. Tax Treatment

Income Tax — Under Model B (Recommended)

Factor	AU	US	NZ	UK
EPM income taxable?	No (loan proceeds)	No (loan proceeds)	No (loan proceeds)	No (loan proceeds)
Portfolio tax	Lender pays	Lender pays	Lender pays (avoids FIF)	Lender pays
CGT on property sale	Exempt (main residence)	Exempt (§121, up to \$500K)	Exempt (no CGT)	Exempt (PPR relief)
Consumer tax compliance	None	None	None	None
Estate/death tax	None	40% above ~\$7M	None	40% above £500K

Retirement Benefit Impact

Region	Benefit	Impact	Severity
AU	Age Pension	⚠️ Significantly reduced — \$640K portfolio exceeds assets test threshold	Critical Disclosure
US	Social Security	✅ Not affected — loan proceeds excluded from MAGI	No Issue
NZ	NZ Super	✅ Not affected — no means test	Marketing Advantage
UK	State Pension	✅ Not affected — NI-based, not means-tested	No Issue
UK	Pension Credit	⚠️ Means-tested — portfolio is countable asset	Disclose
US	Medicaid	⚠️ Asset test — portfolio is countable	Disclose

Best-to-Worst Tax Jurisdictions

New Zealand — No income tax, no estate tax, no pension impact, no CGT

United States (FL, TX, AZ) — No income tax (federal or state), Social Security unaffected

United States (CA, NY) — Same federal treatment but state complexity

Australia — Tax-free income BUT Centrelink impact is a significant negative

United Kingdom — Tax-free income BUT IHT complexity requires careful modelling

5. Estate & Beneficiary Impact

EPM vs Traditional Equity Release

EPM is fundamentally different from traditional equity release.

Traditional ER depletes the estate (compound interest on full balance → NNEG almost certain after 20 years). EPM *grows* the estate — portfolio appreciation outpaces margin accrual.

Factor	Traditional Lifetime Mortgage	FutureProof EPM
Interest mechanism	Compounds on total drawn amount (5-7% AER)	Margin accrues on original LTV only (~3%)
Mortgage after 20 years (£640K)	~£2,050,000	~£778,000
Portfolio at 20 years	N/A (no portfolio)	~£1,200,000
NNEG probability	High	Low
Beneficiary outcome	Little or no inheritance	Significant inheritance likely

Beneficiary Scenarios (20-Year EPM, \$800K Property)

Scenario	Property	Mortgage	Portfolio	Estate	Income Consumed
Best case (strong growth)	\$1,600,000	\$820,000	\$2,100,000	\$2,880,000	\$750,000
Middle case (moderate)	\$1,100,000	\$778,000	\$1,200,000	\$1,522,000	\$600,000
Worst case (crash + NNEG)	\$500,000	\$750,000	\$400,000	\$0	\$350,000

Even in the worst case, the consumer enjoyed \$350,000 in lifetime income and NNEG ensures the estate owes nothing beyond property value.

6. Product Variants

Variant	Target Consumer	Income	Max LTV	Portfolio	Phase
Standard	Retirees seeking regular income	Monthly from day 1	80%	70/30 balanced	Phase 1
Growth	Pre-retirees (55-65)	Deferred 5-10 yrs, then higher	70%	80/20 growth	Phase 2
Flex	Those needing occasional lump sums	On-demand drawdown	60%	60/40 conservative	Phase 2
Protect	Risk-averse, 65+	Monthly with income floor (80% min)	50%	30/70 conservative	Phase 2
Legacy	Estate-focused, other income sources	Reduced (50-70% of Standard)	80%	90/10 aggressive	Phase 3

Growth Variant — Compounding Power

Deferral Period	Portfolio (from \$560K)	Monthly Income	vs Standard
0 years (Standard)	\$560,000	\$1,000	Baseline
5 years	\$785,000	\$1,400	+40%
10 years	\$1,101,000	\$1,900	+90%

7. Security & Data Protection

Current Security Posture

Control	Status	Detail
Rack::Attack rate limiting	✓ Active	Blocklists, throttling, brute force protection
SecureHeaders	✓ Active	HSTS, X-Frame-Options, nosniff, XSS protection
Devise authentication + SSO	✓ Active	SAML, Google OAuth2, Entra ID
PaperTrail audit trail	✓ Active	Model change tracking
reCAPTCHA	✓ Active	Registration protection
Content Security Policy	✓ Strict	No inline scripts/styles

Critical Gaps

Gap	Risk	Priority
InputSanitization disabled on User model	XSS, injection via user registration	Critical
No MFA for admin roles	Account takeover → full platform access	Critical
No field-level encryption (government IDs, credit scores)	PII exposed in plaintext if DB breached	Critical
Devise :lockable not enabled	Unlimited brute force attempts	Critical
Password minimum 6 chars	Weak passwords for financial platform	High
No right to erasure	UK GDPR non-compliance	High
No penetration testing	Unknown vulnerabilities	High

8. Insurance Framework

Four Insurance Layers

Layer	Insurance	Who Pays	Status
Consumer	Buildings insurance (mandatory)	Consumer	Contract covenant
Property	Title insurance	Consumer (one-off)	Required (US), recommended (others)
Portfolio	Hedging (put options, FX forwards, diversification)	Lender (from margin)	Policy required
Platform	PI, cyber, D&O, crime/fidelity	FutureProof	Required for licensing

MPI (Mortgage Protection Insurance) is NOT required for EPM.

The NNEG guarantee provides the same estate protection that MPI would offer. This is a cost saving for consumers versus traditional mortgages.

NNEG as Embedded Put Option

The NNEG is economically equivalent to the lender writing a put option on the property. Pricing factors: LTV ratio, property volatility, consumer age, interest rate environment, property type and location. UK PRA supervisory statement SS3/17 requires Black-Scholes or equivalent pricing methodology.

9. Licensing & Certifications

Cost Summary

~\$400K

One-Time Setup (All Regions)

~\$296K

Annual Ongoing (All Regions)

Critical Path (12 Months)

Month	Milestone
1-2	Company incorporation (each region), director appointments, PI insurance
2-4	Licence applications filed (ACL/AFSL, NMLS + 4 states, FSP/FAP, FCA)
3-6	Staff certifications, SOC 2 Type I prep, Cyber Essentials (UK), AML programs
6-9	AU ACL granted, FL + AZ licences, NZ FAP licence
9-12	AU AFSL, CA + NY licences, UK FCA authorisation, ERC membership, SOC 2 Type I
12+	All licences obtained → launch readiness

Longest Poles

- **UK FCA authorisation:** 6-12 months (start immediately)
- **US New York licence:** 4-6 months (complex state)
- **AU AFSL:** 6-12 months (if investment component classified)

10. Incident Response

Breach Notification Windows

Region	Authority	Window	Threshold
UK	ICO	72 hours	Risk to rights and freedoms
NZ	OPC	~72 hours	Serious harm likely
AU	OAIC	30 days (expedited for serious)	Serious harm likely
US	State AGs	30-90 days (varies)	PII of state residents

Multi-region rule:

The shortest notification window governs. Start ALL notifications within 72 hours (UK ICO deadline). File UK/NZ first, AU/US in parallel.

Financial Calculation Error Protocol

Unique to fintech: if the calculation engine produces incorrect income, the response differs from a standard security incident.

- **Underpaid consumers:** Issue catch-up payment immediately with interest/compensation
- **Overpaid consumers:** Do NOT claw back without legal advice — lender may absorb if amount is small
- **Materiality threshold:** >\$1,000/consumer OR >50 consumers → regulator notification mandatory

11. Currency & Inflation

The FX Problem

EPM portfolios are predominantly USD-denominated (S&P 500, US bonds). Non-US consumers face currency mismatch — portfolio returns in USD, income paid in local currency.

Strategy	Cost	Effectiveness	Recommended For
No hedge	0%	Low — income volatile	—
Income-only hedge	0.2-0.5% p.a.	Medium — protects near-term	All non-US (minimum)
Partial hedge (50%)	0.75-1.5% p.a.	Medium-High	AU, UK
Full hedge	1.5-3% p.a.	High but prohibitively expensive	Not recommended

Inflation & Purchasing Power

Without CPI adjustment, \$1,000/month in 2026 buys only \$554 of goods in 2046 (at 3% inflation). CPI-linked income with a 4% annual cap is sustainable for 25+ years.

Stress Scenario Summary

Scenario	Income	Portfolio	NNEG Risk	Verdict
1970s Stagflation	⚠ Strained	⚠ Strained	✅ Low	Survivable
Japanese Deflation	✅ Floor holds	⚠ Weak	❌ High	Worst case
Currency Crisis	✅ Windfall	✅ Strong (local)	✅ Low	Favourable
Sustained High Inflation	⚠ Rising fast	✅ Strong	✅ Low	Manageable

Deflation is EPM's worst enemy.

Property values fall, NNEG triggers, portfolio weakens. Every other scenario is survivable. NNEG reserves and reinsurance are the defence.

12. Gap Analysis & Priority Actions

Priority 1 — Before Launch

#	Action	Owner	Effort
1	Finalise Model B portfolio ownership structure (legal entity per region)	Legal / Product	2-4 weeks
2	Engage US securities counsel (SEC / Howey test risk assessment)	Legal	Immediate
3	Re-enable InputSanitization + implement MFA + Devise :lockable	Engineering	1-2 days
4	Field-level encryption for government IDs and credit scores	Engineering	2-3 days
5	Centrelink impact estimator in AU quote engine	Engineering	1 week
6	IHT calculator in UK quote engine	Engineering	1 week
7	NNEG clause and beneficiary letter templates in all contracts	Legal	2 weeks
8	File FCA authorisation application (longest lead time)	Compliance	Immediate

Priority 2 — Within 30 Days of Launch

#	Action	Owner
9	Commission penetration test (external firm)	CTO
10	SOC 2 Type I audit initiated	CTO
11	Cyber insurance (\$5M minimum) obtained	Operations
12	Right to erasure endpoint (UK GDPR compliance)	Engineering
13	AML/CFT program implemented (all regions)	Compliance
14	First tabletop incident response drill conducted	CTO

Priority 3 — Within 90 Days

#	Action	Owner
15	CPI data feed integrated (ABS, BLS, Stats NZ, ONS)	Engineering
16	Income-only FX hedge mechanism implemented	Treasury
17	Data retention automation (monthly anonymisation job)	Engineering
18	Portfolio stress testing framework (5 historical scenarios)	Risk
19	Consent management beyond ToS (granular, per-region)	Product

Unresolved Questions

Question	Impact	Owner
Portfolio ownership entity structure per region	Tax, licensing, regulatory in ALL regions	Legal / Tax counsel
AU: Does NNEG affect ATO loan characterisation?	Could reclassify as financial product	AU tax counsel
NZ: Lender entity for portfolio — AU parent or NZ subsidiary?	FIF and corporate tax implications	NZ tax counsel
UK: Does EPM qualify for RNRB?	£175K additional IHT exemption at stake	UK tax counsel
US: TCJA sunset 2026 — estate tax exemption halves	US estate tax relevant for more consumers	US tax counsel
Cross-regional: Who bears FIF/portfolio tax under Model B?	Lender margin calculation	Tax counsel (all)